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# Mitochondrial adaptations to high-volume exercise training are rapidly reversed after a reduction in training volume in human skeletal muscle

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**ABSTRACT: Increased mitochondrial content and respiration have both been reported after exercise training. However, no study has directly compared how different training volumes influence mitochondrial respiration and markers of mitochondrial biogenesis. Ten healthy men performed high-intensity interval cycling during 3 consecutive training phases; 4 wk of normal-volume training (NVT; 3/wk), followed by 20 d of high-volume training (HVT; 2/d) and 2 wk of reduced-volume training (RVT; 5 sessions). Resting biopsy samples (vastus lateralis) were obtained at baseline and after each phase. No mitochondrial parameter changed after NVT. After HVT, mitochondrial respiration and citrate synthase activity (~40–50%), as well as the protein content of electron transport system (ETS) subunits (~10–40%), and that of peroxisome proliferator-activated receptor γ coactivator-1α (PGC-1α), NRF1, mitochondrial transcription factor A (TFAM), PHF20, and p53 (~65–170%) all increased compared to baseline; mitochondrial specific respiration remained unchanged. After RVT, all the mitochondrial parameters measured except citrate synthase activity (~36% above initial) were not significantly different compared to baseline (all $P > 0.05$). Our findings demonstrate that training volume is an important determinant of training-induced mitochondrial adaptations and highlight the rapid reversibility of human skeletal muscle to a reduction in training volume.—Granata, C., Oliveira, R. S. F., Little, J. P., Renner, K., Bishop, D. J. Mitochondrial adaptations to high-volume exercise training are rapidly reversed after a reduction in training volume in human skeletal muscle. FASEB J. 30, 3413–3423 (2016). www.fasebj.org**

**KEY WORDS: mitochondrial biogenesis · mitochondrial respiration · p53 · PGC-1α · PHF20**

Skeletal muscle is a remarkably plastic tissue, rapidly responding to different physiologic stimuli. As a consequence of this plasticity, the repeated physiologic alterations associated with exercise training lead to adaptations such as greater mitochondrial content and respiration (1). However, studies in both mouse (2) and human (3) skeletal muscle have observed an apparent dissociation between training-induced changes in mitochondrial content and respiration, highlighting the importance of measuring and differentiating these 2 parameters (1). Given that mitochondrial adaptations have been associated with improvements in both health (4, 5) and endurance performance (6, 7), while mitochondrial dysfunction has been linked with a wide range of metabolic and neurodegenerative diseases (8, 9), it is important to better understand how mitochondria adapt to the physiologic stress of exercise training.

Recently there has been increased emphasis on manipulating training volume and/or intensity to promote mitochondrial adaptations (1). Both high-volume (6, 10) and high-intensity (7, 11) training have been reported to increase citrate synthase (CS) activity, a valid biomarker of mitochondrial content (12). A correlation has also been observed between training volume and increases in CS activity, suggesting high-volume training (HVT) may be an important determinant of changes in CS activity (1). However, excessive amounts of HVT without adequate recovery (overtraining) can lead to maladaptation (13); for example, studies in overtrained rats have observed reduced CS activity in the red gastrocnemius (14) and

**ABBREVIATIONS:** 10/20k-TT, 10/20-km cycling time trial; AIF, apoptosis-inducing factor; BIOPS, biopsy preservation solution; BM, body mass; CHO, carbohydrate; CI–V, complex I–V; $CIV_{Res}$, reserve capacity of CIV; CS, citrate synthase; DRP1, dynamin-related protein 1; ETS, electron transport system; FCR, flux control ratio; GXT, graded exercise test; HIIT, high-intensity interval training; HVT, high-volume training; NRF1, nuclear respiratory factor 1; NVT, normal-volume training; oxphos, oxidative phosphorylation; PGC-1α, peroxisome proliferator-activated receptor γ coactivator-1α; PHF20, plant homeodomain finger-containing protein 20; RVT, reduced-volume training; SCO2, synthesis of cytochrome *c* oxidase 2; TFAM, mitochondrial transcription factor A; $\dot{V}O_{2Peak}$, peak oxygen uptake; $W_{LT}$, power at lactate threshold; $W_{Peak}$, peak power output

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decreased mitochondrial respiration in the myocardium (15). Despite this evidence, no study has investigated the effects of a period of very high-volume of training on mitochondrial adaptations in human skeletal muscle.

Another important aspect is if, and for how long, mitochondrial adaptations persist when the training stimulus is reduced. Studies indicate that if training intensity is maintained or increased, a reduction in training volume after HVT is accompanied by a series of positive physiologic adaptations (16). For example, the CS activity of highly trained athletes increased by ~18% after 1 wk of reduced-volume, increased-intensity training (17). Conversely, after 1 wk of complete detraining, CS activity was reduced by ~13% (17), and mitochondrial respiration measured in muscle homogenates of elite swimmers decreased by 50% (18). Although these findings highlight the rapid and specific adaptability of skeletal muscle to different training stimuli, no study has directly compared the effects of a period of reduced-volume training (RVT) on both mitochondrial respiration and markers of mitochondrial content.

The underlying molecular mechanisms by which changes in the level of physical activity mediate mitochondrial adaptations are only beginning to be understood. Peroxisome proliferator-activated receptor γ coactivator-1α (PGC-1α) is a transcriptional coactivator regulating mitochondrial biogenesis (19). The PGC-1α action takes place by coactivation of, among others, nuclear respiratory factor 1 (NRF1), which in turn activates mitochondrial transcription factor A (TFAM), providing a link between nuclear and mitochondrial gene expression (20). Similar to PGC-1α, the tumor suppressor p53 is another key transcription factor that modulates mitochondrial biogenesis (19) and can regulate the mitochondrial transcriptional machinery *via* interaction with TFAM (19). The protein content of both PGC-1α and p53 has been shown to increase after exercise training in humans (3, 11). p53 can be up-regulated and stabilized (21) by plant homeodomain finger-containing protein 20 (PHF20), a transcription factor that is also sensitive to exercise training (3). However, the impact of alterations in training volume on the protein content of these transcription factors, as well as their possible association with changes in mitochondrial respiration and content, remain to be determined.

Therefore, the aim of our research was to investigate the effects of 4 wk of normal-volume training (NVT), followed by 3 wk of HVT and then 2 wk of RVT, on skeletal muscle mitochondrial adaptations in young, healthy, moderately trained participants. We hypothesized that markers of mitochondrial content, mitochondrial respiration, and the content of key proteins involved in the regulation of mitochondrial biogenesis would be increased after HVT and these changes would be reversed after RVT. We also expected changes in the protein content of the transcription factors to be associated with changes in both mitochondrial respiration and content.

## MATERIALS AND METHODS

### Participants and ethics approval

Eleven men (20 ± 2 yr; 180 ± 11 cm; 80 ± 14 kg; 45.1 ± 7.6 ml · min$^{-1}$ · kg$^{-1}$) volunteered to take part in this study. Only healthy men aged 18 to 35 yr who were nonsmokers, free of medications before and during the study, moderately trained (*i.e.*, a maximum of 3–4 h · wk$^{-1}$ of light, unstructured aerobic activity for the 6 mo preceding the study), not regularly engaged in cycling-based sports, and not following a structured training program were allowed to take part in this research. After medical screening to rule out cardiovascular, metabolic, and musculoskeletal problems (or any other condition that might have precluded their participation in this study), participants were informed of the study requirements, risks, and benefits before providing written informed consent. Approval for the study's procedures, which conformed to the standards set by the latest revision of the Declaration of Helsinki, was granted by the Victoria University human research ethics committee. Ten participants completed the study, with one participant withdrawing because of time constraints.

### Study design

The study consisted of 3 consecutive training phases: the NVT, HVT, and RVT phases, lasting 4, 3, and 2 wk, respectively. Each training phase was preceded and followed by a testing period in which participants performed a 20-km cycling time trial (20k-TT), a graded exercise test (GXT) (participants were previously familiarized with both tests), and a resting muscle biopsy. The study duration was ~14 wk.

### Testing procedures

Participants were required to refrain from any strenuous physical activity for the 48 h before each performance test (72 h for the biopsy trial), from alcohol and any exercise for 24 h before testing, and from food and caffeine consumption for the 2 h preceding each test. Each type of test was performed at the same time of the day during the course of the entire study to avoid variations caused by circadian rhythm changes.

#### *20k-TT and 10k-TT*

Cycling time trials were performed on an electronically braked cycle ergometer (Velotron, RacerMate, Seattle, WA, USA) after a warm-up involving cycling for 4 min at 66% of the power attained at the lactate threshold ($W_{LT}$), followed by 2 min at $W_{LT}$ and 2 min of rest. During time trials, participants were only allowed access to cadence and completed distance. Heart rate was monitored (Polar-Electro, Kempele, Finland) during all exercise trials and training sessions.

#### *GXT*

A discontinuous GXT was performed on an electronically braked cycle ergometer (Lode Excalibur, v2.0, Groningen, The Netherlands) to determine peak oxygen uptake ($\dot{V}O_{2Peak}$), peak power output ($W_{Peak}$), $W_{LT}$ [using the modified $D_{Max}$ method (22)], and the training intensity for the 3 training phases. The test consisted of 4 min stages at constant power output, interspersed with 30-s rest periods. The test began at 60, 90, or 120 W depending on participants' fitness levels, and was subsequently increased by 30 W every 4 min. Before the test and during the 30-s rest, capillary blood samples were taken from the fingertip for measurement of blood lactate concentration ([La$^{-}$]). Participants were instructed to keep a cadence of >60 rpm and were only allowed access to cadence and elapsed time. The test was stopped when a participant reached volitional exhaustion or cadence dropped below 60 rpm. The $W_{Peak}$ was determined as the power of the last completed stage plus 7.5 W for every additional minute completed.

### Gas analysis during GXT

During the GXT, expired air was continuously analyzed for $O_2$ and $CO_2$ concentrations *via* a gas analyzer (Moxus 2010; AEI Technologies, Pittsburgh, PA, USA), which was calibrated immediately before each test. The ventilometer was calibrated using a 3-L syringe (Hans Rudolph, Shawnee, KS, USA). $\dot{V}O_2$ values were recorded every 15 s, and the average of the 2 highest consecutive 15-s values was recorded as a participant's $\dot{V}O_{2Peak}$.

### Capillary blood sampling

Glass capillary tubes were used to collect ~50 μl of blood at the various time points during the GXT. Capillary blood [$La^-$] was determined using a blood lactate analyzer (YSI 2300 Stat Plus, YSI, Yellow Springs, OH, USA) that was regularly calibrated using precision standards.

### Muscle biopsies

Resting muscle biopsies were performed on the vastus lateralis muscle using a biopsy needle with suction under local anesthesia (1% xylocaine) of the skin and fascia at the following 4 timepoints: before NVT, and after the NVT, HVT, and RVT phases. Muscle samples were cleaned of excess blood, fat, and connective tissue; one portion (10–20 mg) was immediately immersed in ~3 ml of ice-cold biopsy preservation solution (BIOPS) for *in situ* measurements of mitochondrial respiration; the remaining portion was immediately frozen in liquid nitrogen and stored at −80°C for subsequent analyses.

## Training intervention

All training sessions were performed on an electronically braked cycle ergometer (Velotron, RacerMate) and were preceded by an 8 min warm-up (see "20k-TT and 10k-TT"). Participants performed high-intensity interval training (HIIT; 2:1 work-to-rest ratio) at a training intensity set between $W_{LT}$ and $W_{Peak}$ throughout the entire study. This allowed for training volume to be the only parameter manipulated between the 3 training phases. Training intensity was set relative to $W_{LT}$ rather than $W_{Peak}$, as metabolic and cardiac stresses are similar when individuals of differing fitness levels exercise at a percentage of $W_{LT}$, but can vary significantly when training at a percentage of $W_{Peak}$ (23).

### NVT phase

Participants performed HIIT 3 times a week for 4 wk. Training sessions consisted of four to seven 4-min cycling intervals interspersed with a 2-min recovery at 60 W. Training intensities were defined as $[W_{LT} + x(W_{Peak} - W_{LT})]$ (with $x$ equivalent to 0.35, 0.50, 0.65, and 0.75 for wk 1 to 4, respectively).

### HVT phase

Participants performed HIIT twice a day for 20 consecutive days. Training sessions consisted of either five to twelve 4-min intervals at intensities ranging from $[W_{LT} + 0.3(W_{Peak} - W_{LT})]$ to $[W_{LT} + 0.8(W_{Peak} - W_{LT})]$, or eight to twenty-two 2-min intervals at intensities ranging from $[W_{LT} + 0.5(W_{Peak} - W_{LT})]$ to $[W_{LT} + 0.8(W_{Peak} - W_{LT})]$, interspersed with a 1-min recovery at 60 W. Single-session duration increased from 30–35 min to 70–80 min. To monitor participants for signs of overreaching, a 10k-TT was performed before and at regular weekly intervals during this phase. Our intention was that if a participant decreased his 10k-TT performance by more than 10% compared to the initial test (24), then the training load would be decreased to prevent overreaching. However, no participant reduced his endurance performance, and all completed the HVT phase as planned.

### RVT phase

The RVT phase consisted of 5 sessions spread over 14 d. Participants performed four 4-min intervals at an intensity of $[W_{LT} + 0.7(W_{Peak} - W_{LT})]$ during the first session, followed by 4 sessions consisting of five to one 2-min intervals at intensities ranging from $[W_{LT} + 0.8(W_{Peak} - W_{LT})]$ to $[W_{LT} + 0.9(W_{Peak} - W_{LT})]$.

## Physical activity and nutritional control

Participants were required to fill in a food and activity diary to monitor their normal dietary pattern and routine physical activity, which were both maintained throughout the study. To minimize variability in muscle metabolism attributable to diet, participants were provided with a standardized dinner (55 kJ · kg body mass [BM]$^{-1}$, providing 2.1 g carbohydrate [CHO] · kg BM$^{-1}$, 0.3 g fat · kg BM$^{-1}$, and 0.6 g protein · kg BM$^{-1}$) and breakfast (41 kJ · kg BM$^{-1}$, providing 1.8 g CHO · kg BM$^{-1}$, 0.2 g fat · kg BM$^{-1}$, and 0.3 g protein · kg BM$^{-1}$), to be consumed 15 and 3 h before the biopsy trials, respectively. Participants were also required to fill in a nutrition diary recording the last 3 meals before each performance test undertaken during baseline testing and were asked to replicate the same nutritional intake thereafter before the same type of test. To support the HVT, all participants were administered 1 to 2 g · kg BM$^{-1}$ of CHO (Poly-Joule, Nutricia, Macquarie Park, NSW, Australia) during and a further 1 to 2 g · kg BM$^{-1}$ within 1 h after each training session for the entire duration of the HVT phase. The aim was to reach a total daily CHO intake of 10 to 12 g · kg BM$^{-1}$, consistent with current sport nutrition guidelines (25).

## Muscle analyses

### Preparation of whole-muscle lysates

Approximately 10 to 20 mg of frozen muscle was homogenized in an ice-cold lysis buffer (1:20 w/v) containing 50 mM Tris, 150 mM NaCl, 1% Triton X-100, 0.5% sodium deoxycholate, 0.1% SDS, 1 mM EDTA, a protease/phosphatase inhibitor cocktail (Cell Signaling Technology, Danvers, MA, USA), and 1 mM PMSF, adjusted to pH 7.4. Muscle homogenates were rotated at 4°C for 60 min and centrifuged at 15,000 *g* at 4°C for 20 min, and the supernatant was used for immunoblotting and enzyme activity assay. Protein concentration was determined in triplicate using a commercial colorimetric assay (Bio-Rad Protein Assay Kit-II; Bio-Rad, Hercules, CA, USA).

### Immunoblotting

Muscle lysates (10–25 μg) were separated by electrophoresis using SDS-PAGE gels (8–12%), then transferred to PVDF membranes before being blocked for 1 h in 5% skim milk in Tris-buffered saline with 0.1% Tween-20. Membranes were incubated overnight at 4°C with the following primary antibodies: apoptosis-inducing factor (AIF), dynamin-related protein 1 (DRP1), mitofusin 2 (band at ~80 kDa), p53, PHF20 (all Cell Signaling Technology), PGC-1α (EMD Millipore, Billerica, MA, USA), synthesis of cytochrome *c* oxidase (COX) 2 (SCO2) (Santa Cruz Biotechnology, Santa Cruz, CA, USA), NRF1 (band at ~67 kDa), TFAM (band at ~29 kDa), and Total Oxphos (Abcam, Cambridge, MA, USA). Membranes were then incubated at room temperature with the appropriate host species–specific secondary antibody for 90 min before being exposed to a chemiluminescent solution. Three washes with Tris-buffered saline with 0.1% Tween-20 were performed between each step. Protein bands were visualized using a Bio-Rad Versa-Doc imaging system, and bands were quantified using Bio-Rad Quantity One software. An internal standard was loaded in each gel, and each lane was normalized to this value to reduce gel-to-gel variability. Coomassie Blue staining was performed to verify correct loading and equal transfer between lanes (26). Statistical analysis was performed using the raw density data. For graphical purposes, each participant's time point value was normalized to pre-NVT values; therefore, immunoblot graphs are presented as fold change compared to baseline. Moreover, each participant's pre-NVT value was normalized to the mean of the pre-NVT values to obtain a standard deviation. A representative blot for each protein analyzed is presented above the relevant figure.

### *CS activity assay*

CS activity was determined in triplicate on a microtiter plate by adding the following: 5 μl of a 2-mg/ml muscle homogenate, 40 μl of 3 mM acetyl coenzyme A, and 25 μl of 1 mM 5,5′-dithiobis(2-nitrobenzoic acid) in Tris buffer to 165 μl of 100 mM Tris buffer (pH 8.3) kept at 30°C. After addition of 15 μl of 10 mM oxaloacetic acid, the plate was immediately placed in a spectrophotometer at 30°C (xMark-Microplate; Bio-Rad), and after 30 s of linear agitation, absorbance at 412 nm was recorded every 15 s for 3 min. CS activity was reported as mol $\cdot$ h$^{-1}$ $\cdot$ kg protein$^{-1}$.

### *Fiber preparation and high-resolution respirometry*

Muscle fibers were mechanically separated using pointed forceps in ice-cold BIOPS, a biopsy-preserving solution containing (in mM) 2.77 $CaK_2EGTA$, 7.23 $K_2EGTA$, 5.77 $Na_2ATP$, 6.56 $MgCl_2$, 20 taurine, 50 MES, 15 $Na_2$phosphocreatine, 20 imidazole, and 0.5 dithiothreitol adjusted to pH 7.1 (27). The plasma membrane was subsequently permeabilized by gentle agitation for 30 min at 4°C in BIOPS containing 50 μg/ml of saponin and was followed by 3 washes in MiR05, a respiration medium containing (in mM, unless specified) 0.5 EGTA, 3 $MgCl_2$, 60 potassium lactobionate, 20 taurine, 10 $KH_2PO_4$, 20 Hepes, 110 sucrose, and 1 g/L bovine serum albumin essentially fatty acid–free, pH 7.1 (27). Mitochondrial respiration was measured in duplicate (from 3 to 4 mg wet weight of muscle fibers) in MiR05 at 37°C using the high-resolution Oxygraph-2k (Oroboros, Innsbruck, Austria). Oxygen concentration (μM) and flux (pmol $\cdot$ s$^{-1}$ $\cdot$ mg$^{-1}$) were recorded using DatLab software. Reoxygenation by direct syringe injection of $O_2$ was necessary to maintain $O_2$ levels between 275 and 450 μM and to avoid potential oxygen diffusion limitation.

### *Mitochondrial respiration protocol*

A substrate-uncoupler-inhibitor titration protocol was used (27), and the substrate-uncoupler-inhibitor titration sequence was as follows: pyruvate (5 mM) and malate (2 mM) in the absence of adenylates were added for measurement of LEAK respiration (L) through complex I (CI) ($CI_L$). ADP (5 mM) was added for measurement of maximal oxidative phosphorylation (oxphos) capacity (P) through CI ($CI_P$), followed by addition of succinate (10 mM) for measurement of P through CI + complex II (CII) combined ($CI+II_P$). Cytochrome *c* (10 μM) was then added to test for outer mitochondrial membrane integrity, followed by a series of stepwise carbonyl cyanide 4-(trifluoromethoxy) phenylhydrazone titrations (0.75–1.5 μM) for measurement of electron transport system (ETS) capacity (E) through CI+II ($CI+II_E$). Rotenone (0.5 μM), an inhibitor of CI, was added to determine E through CII ($CII_E$), while addition of antimycin A (2.5 μM), an inhibitor of complex III (CIII), allowed measurement and correction of residual oxygen consumption capacity, indicative of nonmitochondrial oxygen consumption. Ascorbate (2 mM) and *N*,*N*,*N*′,*N*′-tetramethyl-*p*-phenylenediamine (0.5 mM), artificial electron donors for complex IV (CIV), were then added to measure E through CIV ($CIV_E$). Respiratory flux control ratios (FCRs) were calculated; the leak control ratio is the quotient of $CI_L$ over $CI+II_E$; the phosphorylation control ratio is the quotient of $CI+II_P$ over $CI+II_E$; the coupling control ratio is the quotient of $CI_L$ over $CI+II_P$ and is equivalent to the inverse respiratory control ratio; the substrate control ratio at constant P is the quotient of $CI_P$ over $CI+II_P$; the reserve capacity of CIV ($CIV_{RES}$) was determined as the quotient of $CI+II_P$ over $CIV_E$.

### Statistical analysis

All values are reported as means ± SD unless specified otherwise. One-way ANOVA, with repeated measures for time, were used to test for main effects. Significant main effects were further analyzed by Tukey's honest significant difference *post hoc* test. SigmaStat software (Jandel Scientific, San Rafael, CA, USA) was used for all statistical analyses. The level of statistical significance was set at $P < 0.05$.

## RESULTS

### Training and performance measurements

Participants completed a significantly higher training volume during the HVT phase (14.7 ± 3.7 MJ) compared to both the NVT and RVT phase (3.6 ± 0.6 and 0.6 ± 0.1 MJ, respectively, both $P < 0.001$); the training volume of the RVT phase was also significantly lower compared to that of the NVT phase ($P = 0.015$) (**Fig. 1**). There were no changes in BM throughout the study (data not shown). Compared to baseline, all performance parameters ($\dot{V}O_{2Peak}$, $W_{LT}$, $W_{Peak}$, 20k-TT) improved after HVT (9–19%, all $P < 0.001$) and remained elevated at the end of the study (8–15%, all $P < 0.05$) (**Table 1**).

### Muscle analyses

After NVT, there were no significant changes in any of the measurements from the muscle analyses.

### *CS activity*

After HVT, CS activity increased by 49.9 ± 49.0% ($P = 0.002$) compared to baseline (38.1 ± 32.4% compared to post-NVT, $P = 0.007$) and did not change significantly after RVT (−5.3 ± 28.2%, $P = 0.645$), where it was still elevated compared to baseline (36.2 ± 47.0%, $P = 0.040$) (**Fig. 2**).

### *Protein content of subunits of the 5 ETS complexes*

After HVT, the protein content of all subunits increased compared to baseline (9–41%, all $P < 0.05$) (**Fig. 3**). With

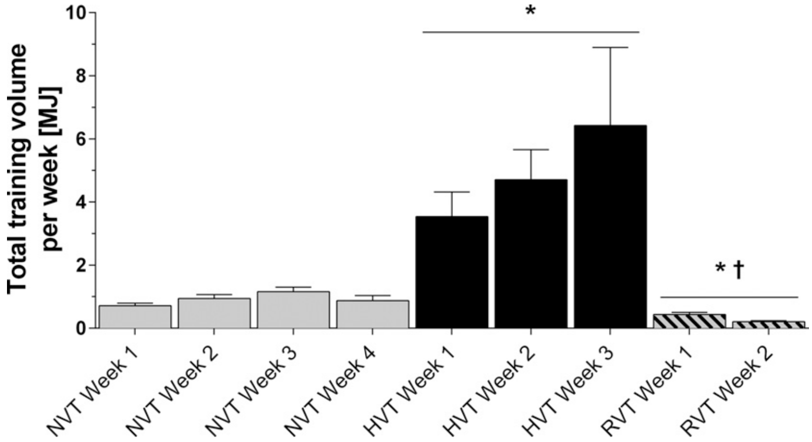


**Figure 1.** Total training volume per week for entire study. Significant differences are indicated. $n$ = 10. All values are means ± SD. *$P < 0.05$ *vs.* NVT, †$P < 0.05$ *vs.* HVT.

the exception of the CIII subunit (6%, $P = 0.138$), the protein content of all subunits also increased as an effect of the HVT phase alone (9–31%, all $P < 0.05$). During the RVT phase, the protein content of the subunits of CI, CII, and CIV was decreased (10–20%, all $P < 0.05$), while that of CIII and CV did not change significantly (−3 to −5%, both $P > 0.05$). However, by the study's end, the protein content of all subunits was not significantly different compared to baseline (4–13%, all $P > 0.05$).

### Mass-specific mitochondrial respiration

There were no significant changes in $CI_L$ throughout the study (all $P > 0.05$) (**Fig. 4*A***). After HVT, measurements of oxphos capacity ($CI_P$ and $CI{+}II_P$) increased by 44 and 41%, respectively (both $P < 0.001$), compared to baseline (47 and 49%, respectively, compared to post-NVT, both $P < 0.001$) and decreased by 14 and 17%, respectively, after RVT (both $P < 0.05$). However, by the study's end, only $CI_P$ was still elevated compared to baseline (24%; $P = 0.034$). After HVT, measurements of ETS capacity ($CI{+}II_E$ and $CII_E$) increased by 41 and 35%, respectively (both $P = 0.002$), compared to baseline (40 and 37%, respectively, compared to post-NVT, both $P < 0.05$) and decreased by 25 and 21%, respectively, after RVT (both $P < 0.05$), where they were not significantly different from baseline (both $P > 0.05$). After HVT, $CIV_E$ increased by 22% ($P = 0.049$) compared to baseline (21% compared to post-NVT, $P = 0.058$) and did not change significantly after RVT (−13%, $P = 0.100$), where it was not significantly different compared to baseline ($P = 0.986$). There was no effect of the addition of cytochrome *c* as a control for outer mitochondrial membrane integrity. FCR changes are shown in **Table 2**.

### Mitochondrial specific respiration (mass-specific mitochondrial respiration normalized by CS activity)

When normalized to CS activity, mitochondrial respiration did not change throughout the study for any of the 6 respiration states (all $P > 0.05$) (Fig. 4*B*).

### Protein content of transcription factors

After HVT, there was a ~70% increase in the protein content of PGC-1α, NRF1, and TFAM (all $P < 0.05$), a 96% increase in PHF20 protein content ($P = 0.022$), and a 168% increase in p53 protein content ($P = 0.002$) (**Fig. 5**). During RVT, there was a significant decrease in the protein content of NRF1 and TFAM (19 and 49%, respectively, both $P < 0.05$). By the study's end, no transcription factor protein content was significantly different from baseline (all $P > 0.05$).

### Protein content of AIF, SCO2, mitofusin 2, and DRP1

The protein content of AIF, SCO2, mitofusin 2, and DRP1 did not change significantly throughout the study (all $P > 0.05$) (**Fig. 6**).

TABLE 1. *Participants' physiologic and endurance performance measurements before and after each training phase*

| Measurement | Pre-NVT | Post-NVT | Post-HVT | Post-RVT |
|---|---|---|---|---|
| $\dot{V}O_{2Peak}$ | 45.1 ± 7.6 | 46.9 ± 7.6 | 52.2 ± 7.7$^{*,\#}$ | 48.9 ± 8.0$^{*,\dagger}$ |
| $W_{LT}$ | 199.9 ± 28.1 | 216.5 ± 30.5* | 233.5 ± 34.9$^{*,\#}$ | 228.0 ± 33.3* |
| $W_{Peak}$ | 265.8 ± 39.0 | 294.0 ± 36.1* | 313.7 ± 40.2$^{*,\#}$ | 302.6 ± 41.7* |
| 20k-TT time | 2235.2 ± 149.2 | 2136.0 ± 95.4* | 2033.2 ± 84.1$^{*,\#}$ | 2047.2 ± 98.8$^{*,\#}$ |

$\dot{V}O_{2Peak}$, peak oxygen uptake (ml · min$^{-1}$ · kg$^{-1}$); $W_{LT}$, power at the lactate threshold (W); $W_{Peak}$, peak power output (W); 20k-TT, 20 km time trial (s). Significant differences are indicated. $n$ = 10. All values are means ± SD. *$P < 0.05$ *vs.* pre-NVT, #$P < 0.05$ *vs.* post-NVT, †$P < 0.05$ *vs.* post-HVT.

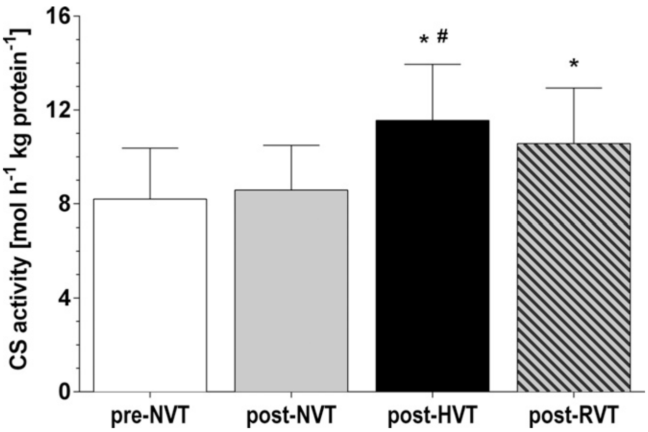


**Figure 2.** CS activity expressed per kilogram of protein. CS activity was measured in whole-muscle lysates prepared from muscle biopsy samples (vastus lateralis) obtained at rest at baseline (pre-NVT) and after the NVT, HVT, and RVT phases. Significant differences are indicated. $n$ = 10. All values are means ± SD. *$P < 0.05$ *vs.* pre-NVT, $^{\#}P < 0.05$ *vs.* post-NVT.

## DISCUSSION

Three weeks of HVT improved endurance performance, mitochondrial content, and mitochondrial respiration. These improvements were accompanied by an increase in the protein content of key transcription factors regulating mitochondrial biogenesis. A striking finding of our research was that 2 wk of RVT was sufficient to reverse the majority of the skeletal muscle mitochondrial gains obtained during the HVT phase, with most parameters returning to values that were not significantly different from baseline. Our results highlight that training volume is an important determinant of changes in mitochondrial content and that human skeletal muscle is a remarkably plastic tissue, rapidly adapting to both increases and decreases in training volume.

### NVT phase

The NVT phase resulted in no significant changes in any of the measured muscle parameters and in a small improvement in $W_{LT}$, $W_{Peak}$, and 20k-TT time. Results from the NVT phase will not be discussed further because these represent a subset of a training group whose results have been published elsewhere (3).

### Performance

Indicators of endurance performance ($\dot{V}O_{2Peak}$, $W_{Peak}$, $W_{LT}$, and 20k-TT time) improved during HVT (~5–12%), where all participants registered peak values for the study. These findings are in contrast to previous research showing a reduction in markers of endurance performance after 2 wk of HVT (24). These discrepancies may relate to the CHO supplementation during the HVT reducing or delaying the symptoms of overreaching (28). Furthermore, in the study by Halson *et al.* (24), the increase in training volume was accompanied by an overall increase in training intensity. In our study, training intensity did not change to allow us to isolate the effect of manipulating training volume. Therefore, our research indicates that when CHO levels are kept high, moderately trained individuals are able to positively adapt to a period of HVT (2 HIIT sessions per day, each lasting up to 80 min, for 20 consecutive days) when training intensity is maintained. As a consequence of the positive adaptations that followed HVT, the RVT phase, which was associated with an ~95% reduction in training volume, resulted in a modest loss of whole-body exercise

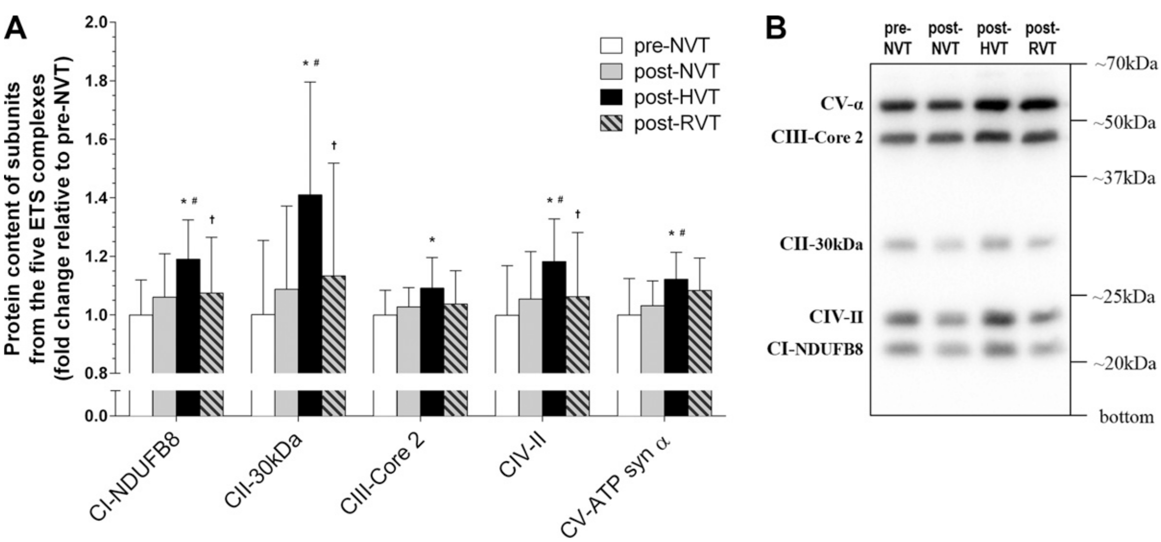


**Figure 3.** *A*) Fold change compared to baseline of the protein content of subunits from the 5 complexes of the ETS, CI to CV, in whole-muscle lysates prepared from skeletal muscle biopsy (vastus lateralis). Biopsy samples were obtained at rest at baseline (pre-NVT), and after the NVT, HVT, and RVT phases. *B*) Representative immunoblot of the subunits of the 5 ETS complexes. Significant differences are indicated. $n$ = 10. All values are means ± SD. *$P < 0.05$ *vs.* pre-NVT, $^{\#}P < 0.05$ *vs.* post-NVT, $^{\dagger}P < 0.05$ *vs.* post-HVT.

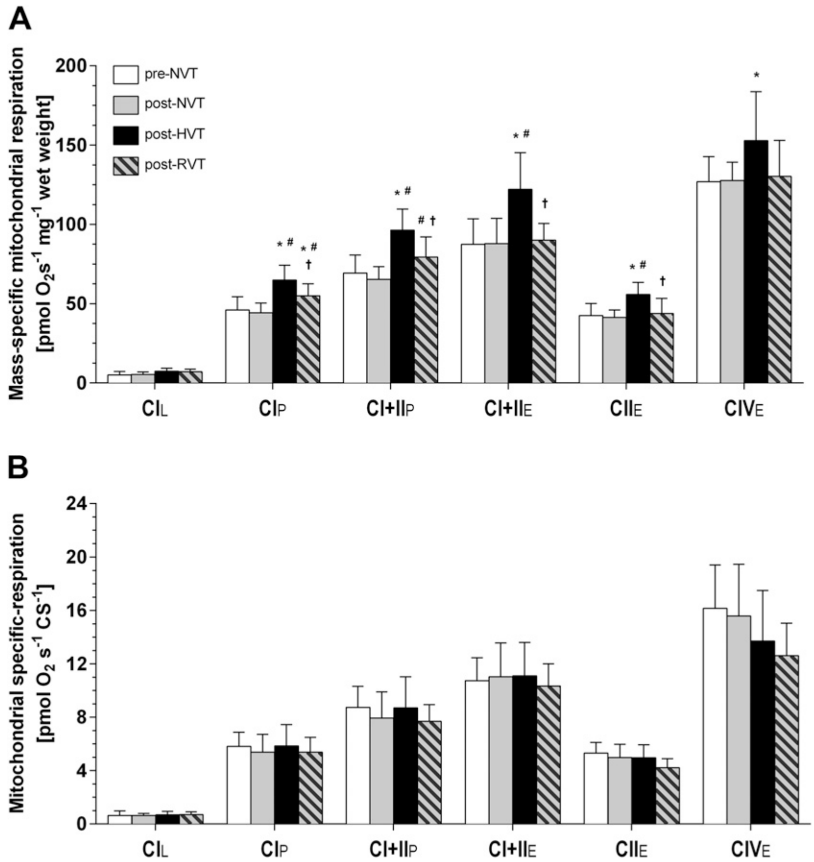


**Figure 4.** Mitochondrial respiration measurements at different coupling control states. *A*) Mass-specific mitochondrial respiration. *B*) Mitochondrial specific respiration (obtained by normalizing mass-specific mitochondrial respiration by CS activity expressed per kilogram of protein). Respiration was measured in permeabilized fibers prepared from muscle biopsy samples (vastus lateralis) obtained at rest at baseline (pre-NVT), and after the NVT, HVT, and RVT phases. Respiration values are as follows: $CI_L$, leak respiration state (L) in absence of adenylates and limitation of flux by electron input through CI; $CI_P$, maximal oxphos state (P) with saturating levels of ADP and limitation of flux by electron input through CI; $CI+II_P$, P with saturating levels of ADP and limitation of flux by convergent electron input through CI+II; $CI+II_E$, maximal ETS capacity (E) with saturating levels of ADP and limitation of flux by convergent electron input through CI+II; $CII_E$, E with saturating levels of ADP and limitation of flux by electron input through CII; $CIV_E$, maximal CIV noncoupled respiration with saturating levels of ADP and limitation of flux by electron input through CIV. Significant differences are indicated. $n$ = 10. All values are means ± SD. $^*P < 0.05$ *vs.* pre-NVT, $^{\#}P < 0.05$ *vs.* post-NVT, $^{\dagger}P < 0.05$ *vs.* post-HVT.

performance (~1–6%). This is consistent with previous research reporting that unless training intensity is increased, a subsequent reduction in training volume does not improve endurance performance (17).

## Markers of mitochondrial content

It has been suggested in a recent review that there is a relationship between training volume and changes in CS activity (1); however, this hypothesis has not been tested within the one study. By directly comparing 3 different training volumes, our findings confirm this relationship. The large increase in CS activity after HVT (50%) is consistent with that of previous research that uses similar training volumes (6, 10), and it agrees well with the predicted relationship between training volume and training-induced changes in CS activity (1). After RVT, CS activity did not change significantly (−5%), consistent with the lack of change reported after 1 wk of reduced training volume at a maintained training intensity in the vastus lateralis muscle of elite runners (17). Our findings also agree with the nonsignificant decrease in CS activity in the vastus lateralis muscle reported after a 3-wk reduction in training volume in recreational runners (29) or after a 4 wk training volume reduction in well-trained endurance athletes (30). Conversely, complete detraining has been associated with significant decreases in CS activity, ranging

TABLE 2. *Respiratory FCRs*

| FCR | Pre-NVT | Post-NVT | Post-HVT | Post-RVT |
|---|---|---|---|---|
| LCR (L/E) | 0.04 ± 0.02 | 0.07 ± 0.04 | 0.06 ± 0.02 | 0.08 ± 0.01 |
| PCR (P/E) | 0.81 ± 0.07 | 0.74 ± 0.03 | 0.77 ± 0.04 | 0.81 ± 0.03 |
| Inv-RCR (L/P) | 0.07 ± 0.04 | 0.09 ± 0.04 | 0.08 ± 0.02 | 0.09 ± 0.02 |
| SCR (constant P) | 0.66 ± 0.05 | 0.68 ± 0.04 | 0.67 ± 0.03 | 0.70 ± 0.05 |
| $CIV_{RES}$ | 0.55 ± 0.05 | 0.51 ± 0.08 | $0.64 \pm 0.09^{*,\#}$ | $0.62 \pm 0.08^{\#}$ |

L, leak respiration; P, oxphos capacity; E, ETS capacity; LCR, leak control ratio ($CI_L/CI+II_E$); PCR, phosphorylation control ratio ($CI+II_P/CI+II_E$); Inv-RCR, inverse of respiratory control ratio ($CI_L/CI+II_P$); SCR, substrate control ratio at constant P ($CI_P/CI+II_P$); $CIV_{RES}$, CIV reserve capacity ($CI+II_P/CIV_E$); CI, electron input through CI; CI+II, convergent electron input through CI and CII; CIV, electron input through CIV. FCR were calculated from mass-specific mitochondrial respiration measurements in permeabilized muscle fibers (vastus lateralis), obtained from muscle biopsy samples taken at baseline and after each training phase. Significant differences are indicated. $n$ = 10. All values are means ± SD. $^*P < 0.05$ *vs.* pre-NVT, $^{\#}P < 0.05$ *vs.* post-NVT.

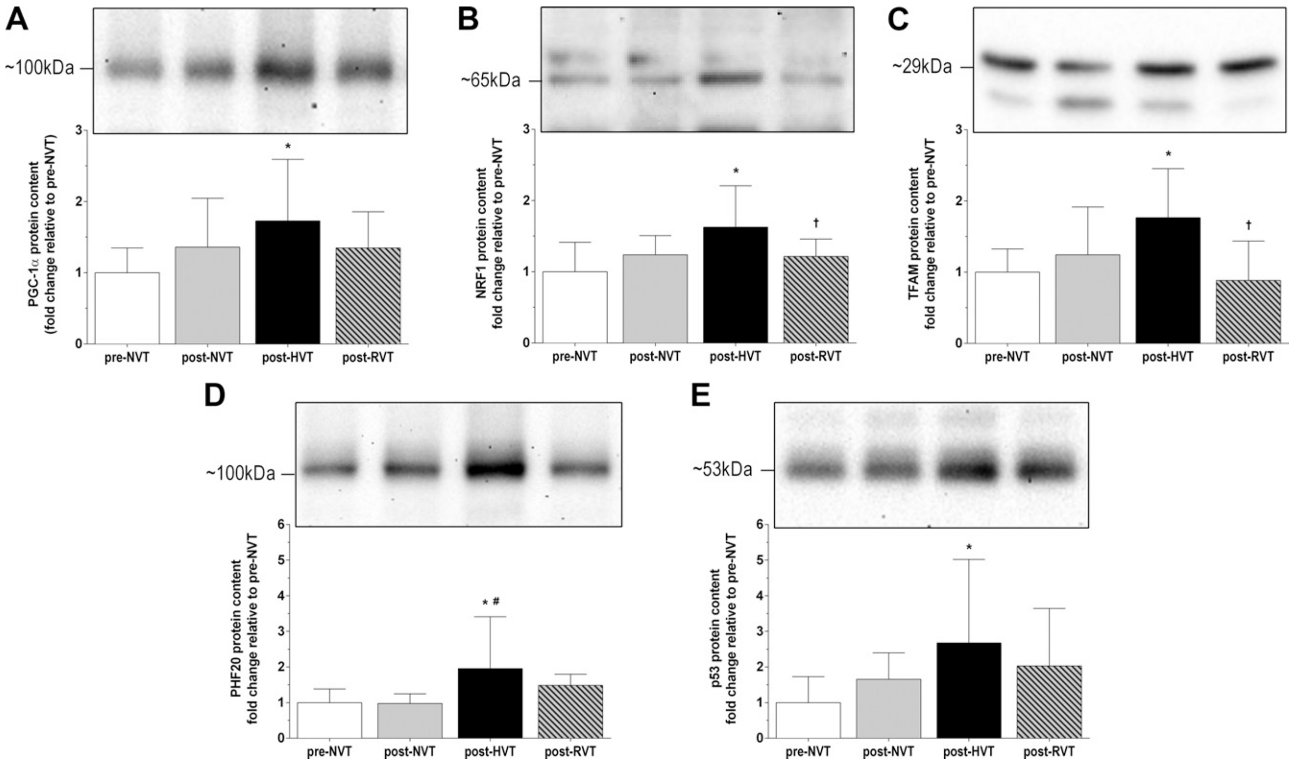


**Figure 5.** Fold change compared to baseline of the protein content of PGC-1α (*A*), NRF1 (*B*), TFAM (*C*), PHF20 (*D*), and p53 (*E*) in whole-muscle lysates prepared from skeletal muscle biopsy (vastus lateralis). Biopsy samples were obtained at rest at baseline (pre-NVT), and after the NVT, HVT, and RVT phases. Representative immunoblots are presented above each target protein. Significant differences are indicated. $n = 10$. All values are means ± SD. $^{*}P < 0.05$ *vs.* pre-NVT, $^{\#}P < 0.05$ *vs.* post-NVT, $^{\dagger}P < 0.05$ *vs.* post-HVT.

from 10% in 1 wk (17) to 30% after 10 d (31), and up to 45% in 3 wk (32). At the end of our study, CS activity was still elevated compared to baseline (36%), suggesting that despite an ~95% reduction in training volume, maintenance of a minimal amount of training may help preserve CS activity and avoid the loss associated with complete detraining.

After HVT, and similar to CS activity, we observed a 9–41% increase in the protein content of subunits of the ETS complexes, another valid biomarker of mitochondrial content, even though the strength of this relationship is weaker than that for CS activity and varies between complexes (12). We are not aware of previous research investigating changes in the content of these complexes after HVT (or RVT). In contrast to CS activity, RVT induced a 10–20% significant decrease in the content of the subunits of 3 of the ETS complexes. Moreover, by the study's end, the content of all 5 subunits was not significantly different from baseline. This suggests the training volume during RVT did not provide a sufficient stimulus to maintain the increase in mitochondrial protein content observed after HVT. It may also indicate that the protein content of the ETS complexes is more sensitive to a reduction in training volume than CS activity, an observation that may relate to the different half-lives of mitochondrial complexes and CS (33, 34). Taken together, these findings highlight the plasticity and rapid adaptability of human skeletal muscle to different training stimuli and further support the hypothesis that training volume is an important determinant of changes in mitochondrial content (1).

## Mitochondrial respiration

Although previous research observed that training at an intensity between $W_{LT}$ and $W_{Peak}$ was not sufficient to improve mitochondrial respiration (3), results from the present study indicate this intensity is sufficient to increase mitochondrial respiration (possibly *via* increased mitochondrial content) when combined with HVT. After HVT, all mass-specific mitochondrial respiration states in the presence of ADP were increased (20–50%) compared to both baseline and post-NVT values. Conversely, mitochondrial specific respiration, obtained by normalizing mass-specific mitochondrial respiration by CS activity, did not change throughout the study. This suggests that changes in mass-specific mitochondrial respiration were mainly the result of changes in mitochondrial content. However, all-out training intensity has been shown to enhance mass-specific mitochondrial respiration without changes in mitochondrial content (3). This suggests that mitochondrial respiration may be improved by either training at maximal intensity, which leads to greater mitochondrial-specific respiration (3), and/or by increasing training volume, which leads to greater mitochondrial content and no apparent changes in mitochondrial specific respiration.

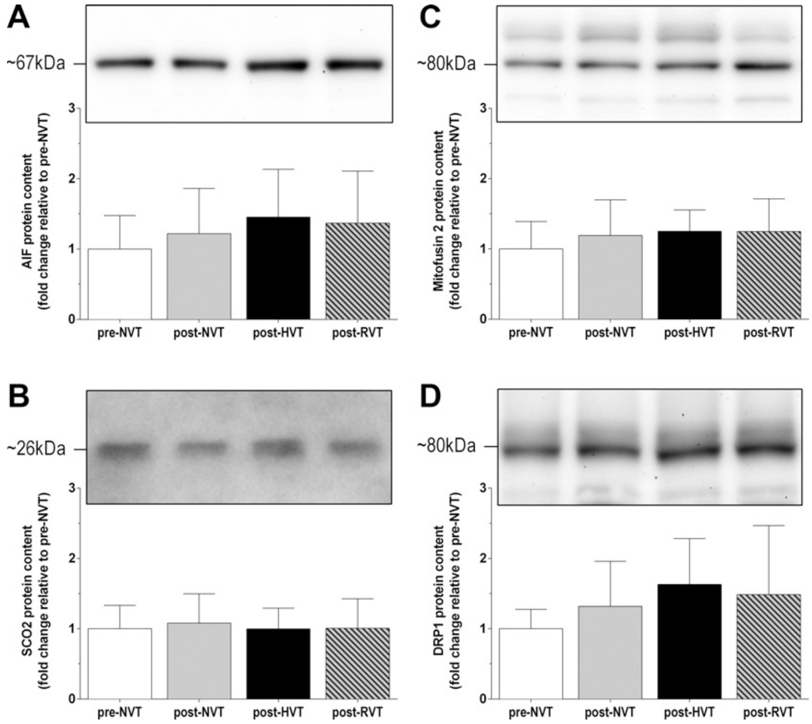


**Figure 6.** Fold change compared to baseline of the protein content of AIF (*A*), SCO2 (*B*), mitofusin 2 (*C*), and DRP1 (*D*) in whole-muscle lysates prepared from skeletal muscle biopsy (vastus lateralis) sample. Biopsy samples were obtained at rest at baseline (pre-NVT), and after the NVT, HVT, and RVT phases. Representative immunoblots are presented above each target protein. $n$ = 10. All values are means ± SD.

It has been hypothesized that training intensity is a more important determinant of changes in mitochondrial respiration than training volume (1). In the present study, we observed a similar increase in $CI_P$ to the interval training group in Daussin *et al.* (35), despite a ~3-fold greater training volume. Increases of ~20–25% in $CI{+}II_P$ (3, 7) were also reported after training volumes equivalent to <10% of those used during HVT, which increased $CI{+}II_P$ by 49%. Although minor methodological variations and differences in participants' fitness levels may have influenced the outcome, these results seem to support the observation that training volume is not the primary determinant of changes in mitochondrial respiration.

After the RVT phase, we observed an ~15–25% decrease in all mass-specific mitochondrial respiration values except $CI_L$. No study to date has investigated mitochondrial respiration in permeabilized fibers after RVT. Nevertheless, mitochondrial respiration measured in deltoid muscle homogenates of elite swimmers was decreased by 50% after 1 wk of detraining (18). The greater decrease reported by Costill *et al.* (18) may relate to the different muscle investigated, training status, the use of detraining *vs.* RVT, and/or the muscle preparation. However, the ~15–25% decrease in mass-specific mitochondrial respiration in the present study is comparable with the 12–28% decrease in mitochondrial ATP production rate after 3 wk of detraining in active men (36). By the end of our 14-wk study, only $CI_P$ was significantly greater compared to baseline, whereas all other respiration states were not significantly different from baseline despite the large improvements after HVT. These findings emphasize the rapid reversibility of mitochondrial respiration gains after a period of RVT and the importance of maintaining the training stimulus to preserve training-induced mitochondrial adaptations.

Analysis of the FCRs showed that there was no change in the index of uncoupling (leak control ratio), the limitation of oxphos capacity by the phosphorylation system (phosphorylation control ratio), the degree of coupling between oxidation and phosphorylation (inverse respiratory control ratio), and substrate control (substrate control ratio) throughout the study. These findings are consistent with the lack of change in mitochondrial specific respiration and seem to indicate mitochondrial quality remained unaltered and that changes in mass-specific mitochondrial respiration were related to an increase in mitochondrial content. $CIV_{RES}$, however, was significantly elevated after HVT compared to baseline. Because CIV oxidative capacity measured in permeabilized fibers is in excess of $CI{+}II_P$ (27), the smaller increase in $CIV_E$ compared to $CI{+}II_P$ after HVT may be due to this overcapacity and the ability of CIV to already sustain a higher overall mitochondrial respiration.

## Mitochondrial proteins and transcription factors

To our knowledge, no study has investigated changes in the protein content of PGC-1α, p53, NRF1, PHF20, and TFAM [5 transcription factors involved in the regulation of mitochondrial biogenesis (19–21)] after HVT. After HVT, the protein content of these transcription factors increased by ~65–170%. The similar and concomitant increase in PGC-1α, NRF1, and TFAM protein content is consistent with the changes reported after 4 d of contractile activity in muscle cells (37) and with the role of the NRF1-TFAM axis as a link between nuclear and mitochondrial gene expression (19, 20). It has previously been hypothesized that existing NRF1 protein is sufficient to meet the cellular needs (38). However, increased NRF1 protein content after HVT raises the possibility that existing NRF1 protein may not be sufficient to satisfy the cellular requirements when training volume is greatly increased. After HVT, we also observed a concomitant increase in mass-specific mitochondrial respiration and the protein content of PGC-1$\alpha$, p53, and PHF20, as previously observed (3, 11). These results are consistent with the role of PGC-1$\alpha$ and p53 as modulators of mitochondrial respiration (3, 19). Although PGC-1$\alpha$, p53, and PHF20 protein content did not change significantly after NVT (3×/wk for 4 wk), our results indicate that when training volume is markedly increased, as during HVT, the training intensity associated with the HIIT in the present study is sufficient to enhance the protein content of these transcription factors in healthy men.

By the study's end, the protein content of all 5 transcription factors decreased to a level that was not significantly different from baseline. Although only TFAM and NRF1 decreased significantly during RVT, this is the first study to show that these transcription factors are sensitive to short-term reductions in training volume. The rapid changes in the content of these transcription factors indicate that even at the cellular level, human skeletal muscle quickly responds to fluctuations in the training stimulus and rapidly adapts to the new metabolic and energy requirements. Moreover, while training at different intensities does not influence changes in TFAM protein content or in mitochondrial content (3), the different volumes of training used in the present study resulted in significant changes in these parameters. This may suggest that changes in TFAM protein content may be related to training volume and may be associated with changes in mitochondrial content. This would also be consistent with the role of TFAM as a key factor regulating the mitochondrial transcriptional machinery (19, 20). Further research is needed to investigate these hypotheses.

p53 has been shown to modulate mitochondrial respiration and mitochondrial remodeling through transcriptional regulation of, among others, SCO2, AIF, mitofusin 2, and DRP1 (19). However, despite the changes in p53 protein content, we report no change in the protein content of these downstream targets of p53 after both HVT and RVT. The lack of change in SCO2 and AIF protein content is consistent with previous findings in human skeletal muscle showing that these targets did not change after exercise training (3, 39). There are contrasting results in regard to changes in the mitochondrial remodeling proteins mitofusin 2 and DRP1 after different types of training (3, 11). Further research is needed to better determine the effects of different types of exercise on these proteins.

## CONCLUSIONS

The HVT phase (40 HIIT sessions lasting up to 80 min each in 20 d) increased whole-body exercise performance, suggesting that when CHO levels are kept high, moderately trained individuals can positively adapt to 3 wk of high-volume, high-intensity training. These positive adaptations were associated with an increase in markers of mitochondrial content, suggesting that training volume may be an important determinant of changes in mitochondrial content. After RVT, endurance performance was maintained, but all parameters measured in the muscle were not significantly different from before the study (except $CI_P$ and CS activity). This suggests that maximal mass-specific mitochondrial respiration may be reversed more rapidly than mitochondrial content (as assessed by CS activity), providing further evidence that these 2 parameters are not necessarily associated. These findings emphasize the rapid reversibility of mitochondrial adaptations after a ~95% reduction in training volume, and they also emphasize the importance of maintaining the training stimulus to preserve previous skeletal muscle remodeling. At the cellular level, changes in the protein content of key transcription factors regulating mitochondrial biogenesis are also modulated by changes in training volume. The findings from the present study provide valuable information to better design exercise prescriptions, suggesting that training volume is an important tool to induce mitochondrial adaptations, and that periods of RVT, unless managed carefully, may lead to a decrease in both mitochondrial content and mitochondrial function. FJ

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## AUTHOR CONTRIBUTIONS

C. Granata and D. J. Bishop designed the research; C. Granata and R. S. F. Oliveira conducted the research; C. Granata, R. S. F. Oliveira, J. P. Little, K. Renner, and D. J. Bishop analyzed and interpreted data; C. Granata wrote the manuscript; C. Granata, R. S. F. Oliveira, J. P. Little, K. Renner, and D. J. Bishop critically revised and contributed to the manuscript; C. Granata and D. J. Bishop have primary responsibility for final content; and all authors have read and approved the final manuscript.

## REFERENCES

1. Bishop, D. J., Granata, C., and Eynon, N. (2014) Can we optimise the exercise training prescription to maximise improvements in mitochondria function and content? *Biochim. Biophys. Acta* **1840**, 1266–1275
2. Rowe, G. C., Patten, I. S., Zsengeller, Z. K., El-Khoury, R., Okutsu, M., Bampoh, S., Koulisis, N., Farrell, C., Hirshman, M. F., Yan, Z., Goodyear, L. J., Rustin, P., and Arany, Z. (2013) Disconnecting mitochondrial content from respiratory chain capacity in PGC-1-deficient skeletal muscle. *Cell Rep.* **3**, 1449–1456
3. Granata, C., Oliveira, R. S., Little, J. P., Renner, K., and Bishop, D. J. (2016) Training intensity modulates changes in PGC-1α and p53 protein content and mitochondrial respiration, but not markers of mitochondrial content in human skeletal muscle. *FASEB J.* **30**, 959–970

4. Gibala, M. J., Little, J. P., Macdonald, M. J., and Hawley, J. A. (2012) Physiological adaptations to low-volume, high-intensity interval training in health and disease. *J. Physiol.* **590**, 1077–1084
5. Meex, R. C. R., Schrauwen-Hinderling, V. B., Moonen-Kornips, E., Schaart, G., Mensink, M., Phielix, E., van de Weijer, T., Sels, J. P., Schrauwen, P., and Hesselink, M. K. C. (2010) Restoration of muscle mitochondrial function and metabolic flexibility in type 2 diabetes by exercise training is paralleled by increased myocellular fat storage and improved insulin sensitivity. *Diabetes* **59**, 572–579
6. Tonkonogi, M., Walsh, B., Svensson, M., and Sahlin, K. (2000) Mitochondrial function and antioxidative defence in human muscle: effects of endurance training and oxidative stress. *J. Physiol.* **528**, 379–388
7. Vincent, G., Lamon, S., Gant, N., Vincent, P. J., MacDonald, J. R., Markworth, J. F., Edge, J. A., and Hickey, A. J. (2015) Changes in mitochondrial function and mitochondria associated protein expression in response to 2-weeks of high intensity interval training. *Front. Physiol.* **6**, 51
8. Luft, R. (1994) The development of mitochondrial medicine. *Proc. Natl. Acad. Sci. USA* **91**, 8731–8738
9. Wang, C. H., Wang, C. C., and Wei, Y. H. (2010) Mitochondrial dysfunction in insulin insensitivity: implication of mitochondrial role in type 2 diabetes. *Ann. N. Y. Acad. Sci.* 1201, 157–165
10. Jeppesen, J., Jordy, A. B., Sjøberg, K. A., Füllekrug, J., Stahl, A., Nybo, L., and Kiens, B. (2012) Enhanced fatty acid oxidation and FATP4 protein expression after endurance exercise training in human skeletal muscle. *PLoS One* **7**, e29391
11. Perry, C. G. R., Lally, J., Holloway, G. P., Heigenhauser, G. J. F., Bonen, A., and Spriet, L. L. (2010) Repeated transient mRNA bursts precede increases in transcriptional and mitochondrial proteins during training in human skeletal muscle. *J. Physiol.* **588**, 4795–4810
12. Larsen, S., Nielsen, J., Hansen, C. N., Nielsen, L. B., Wibrand, F., Stride, N., Schroder, H. D., Boushel, R., Helge, J. W., Dela, F., and Hey-Mogensen, M. (2012) Biomarkers of mitochondrial content in skeletal muscle of healthy young human subjects. *J. Physiol.* **590**, 3349–3360
13. Lehmann, M. J., Lormes, W., Opitz-Gress, A., Steinacker, J. M., Netzer, N., Foster, C., and Gastmann, U. (1997) Training and overtraining: an overview and experimental results in endurance sports. *J. Sports Med. Phys. Fitness* **37**, 7–17
14. Ferraresso, R. L. P., Buscariolli De Oliveira, R., MacEdo, D. V., Alessandro Soares Nunes, L., Brenzikofer, R., Damas, D., and Hohl, R. (2012) Interaction between overtraining and the interindividual variability may (not) trigger muscle oxidative stress and cardiomyocyte apoptosis in rats. *Oxidative Med. Cell. Longevity* 935483
15. Kadaja, L., Eimre, M., Paju, K., Roosimaa, M., Põdramägi, T., Kaasik, P., Pehme, A., Orlova, E., Mudist, M., Peet, N., Piirsoo, A., Seene, T., Gellerich, F. N., and Seppet, E. K. (2010) Impaired oxidative phosphorylation in overtrained rat myocardium. *Exp. Clin. Cardiol.* **15**, e116–e127
16. Mujika, I., Padilla, S., Pyne, D., and Busso, T. (2004) Physiological changes associated with the pre-event taper in athletes. *Sports Med.* **34**, 891–927
17. Shepley, B., MacDougall, J. D., Cipriano, N., Sutton, J. R., Tarnopolsky, M. A., and Coates, G. (1992) Physiological effects of tapering in highly trained athletes. *J. Appl. Physiol.* **72**, 706–711
18. Costill, D. L., Fink, W. J., Hargreaves, M., King, D. S., Thomas, R., and Fielding, R. (1985) Metabolic characteristics of skeletal muscle during detraining from competitive swimming. *Med. Sci. Sports Exerc.* **17**, 339–343
19. Saleem, A., Carter, H. N., Iqbal, S., and Hood, D. A. (2011) Role of p53 within the regulatory network controlling muscle mitochondrial biogenesis. *Exerc. Sport Sci. Rev.* **39**, 199–205
20. Virbasius, J. V., and Scarpulla, R. C. (1994) Activation of the human mitochondrial transcription factor A gene by nuclear respiratory factors: a potential regulatory link between nuclear and mitochondrial gene expression in organelle biogenesis. *Proc. Natl. Acad. Sci. USA* **91**, 1309–1313
21. Cui, G., Park, S., Badeaux, A. I., Kim, D., Lee, J., Thompson, J. R., Yan, F., Kaneko, S., Yuan, Z., Botuyan, M. V., Bedford, M. T., Cheng, J. Q., and Mer, G. (2012) PHF20 is an effector protein of p53 double lysine methylation that stabilizes and activates p53. *Nat. Struct. Mol. Biol.* **19**, 916–924
22. Bishop, D., Jenkins, D. G., McEniery, M., and Carey, M. F. (2000) Relationship between plasma lactate parameters and muscle characteristics in female cyclists. *Med. Sci. Sports Exerc.* **32**, 1088–1093
23. Baldwin, J., Snow, R. J., and Febbraio, M. A. (2000) Effect of training status and relative exercise intensity on physiological responses in men. *Med. Sci. Sports Exerc.* **32**, 1648–1654
24. Halson, S. L., Bridge, M. W., Meeusen, R., Busschaert, B., Gleeson, M., Jones, D. A., and Jeukendrup, A. E. (2002) Time course of performance changes and fatigue markers during intensified training in trained cyclists. *J. Appl. Physiol.* **93**, 947–956
25. Bartlett, J. D., Hawley, J. A., and Morton, J. P. (2015) Carbohydrate availability and exercise training adaptation: too much of a good thing? *Eur. J. Sport Sci.* **15**, 3–12
26. Welinder, C., and Ekblad, L. (2011) Coomassie staining as loading control in Western blot analysis. *J. Proteome Res.* **10**, 1416–1419
27. Pesta, D., and Gnaiger, E. (2012) High-resolution respirometry: oxphos protocols for human cells and permeabilized fibers from small biopsies of human muscle. In *Mitochondrial Bioenergetics: Methods and Protocols* (Palmeira, C. M., Moreno, A. J., eds.), vol. 810, pp. 25–58, Springer Science+Business Media, New York
28. Halson, S. L., Lancaster, G. I., Achten, J., Gleeson, M., and Jeukendrup, A. E. (2004) Effects of carbohydrate supplementation on performance and carbohydrate oxidation after intensified cycling training. *J. Appl. Physiol.* **97**, 1245–1253
29. Luden, N., Hayes, E., Minchev, K., Louis, E., Raue, U., Conley, T., and Trappe, S. (2012) Skeletal muscle plasticity with marathon training in novice runners. *Scand. J. Med. Sci. Sports* **22**, 662–670
30. Madsen, K., Pedersen, P. K., Djurhuus, M. S., and Klitgaard, N. A. (1993) Effects of detraining on endurance capacity and metabolic changes during prolonged exhaustive exercise. *J. Appl. Physiol.* **75**, 1444–1451
31. McCoy, M., Proietto, J., and Hargreves, M. (1994) Effect of detraining on GLUT-4 protein in human skeletal muscle. *J. Appl. Physiol.* **77**, 1532–1536
32. Moore, R. L., Thacker, E. M., Kelley, G. A., Musch, T. I., Sinoway, L. I., Foster, V. L., and Dickinson, A. L. (1987) Effect of training/detraining on submaximal exercise responses in humans. *J. Appl. Physiol.* **63**, 1719–1724
33. Booth, F. W., and Holloszy, J. O. (1977) Cytochrome *c* turnover in rat skeletal muscles. *J. Biol. Chem.* **252**, 416–419
34. Wright, D. C., Han, D. H., Garcia-Roves, P. M., Geiger, P. C., Jones, T. E., and Holloszy, J. O. (2007) Exercise-induced mitochondrial biogenesis begins before the increase in muscle PGC-1α expression. *J. Biol. Chem.* **282**, 194–199
35. Daussin, F. N., Zoll, J., Dufour, S. P., Ponsot, E., Lonsdorfer-Wolf, E., Doutreleau, S., Mettauer, B., Piquard, F., Geny, B., and Richard, R. (2008) Effect of interval versus continuous training on cardiorespiratory and mitochondrial functions: relationship to aerobic performance improvements in sedentary subjects. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **295**, R264–R272
36. Wibom, R., Hultman, E., Johansson, M., Matherei, K., Constantin-Teodosiu, D., and Schantz, P. G. (1992) Adaptation of mitochondrial ATP production in human skeletal muscle to endurance training and detraining. *J. Appl. Physiol.* **73**, 2004–2010
37. Irrcher, I., Adhihetty, P. J., Sheehan, T., Joseph, A. M., and Hood, D. A. (2003) PPARgamma coactivator-1α expression during thyroid hormone– and contractile activity–induced mitochondrial adaptations. *Am. J. Physiol. Cell Physiol.* **284**, C1669–C1677
38. Pilegaard, H., Saltin, B., and Neufer, P. D. (2003) Exercise induces transient transcriptional activation of the PGC-1α gene in human skeletal muscle. *J. Physiol.* **546**, 851–858
39. Adhihetty, P. J., Taivassalo, T., Haller, R. G., Walkinshaw, D. R., and Hood, D. A. (2007) The effect of training on the expression of mitochondrial biogenesis- and apoptosis-related proteins in skeletal muscle of patients with mtDNA defects. *Am. J. Physiol. Endocrinol. Metab.* **293**, E672–E680

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# Mitochondrial adaptations to high-volume exercise training are rapidly reversed after a reduction in training volume in human skeletal muscle

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